Assignment Report

SFWRENG 3S03 (2026W)

Assignment 3

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GitHub URL

`https://github.com/hydrowoxy/3S03-A3`

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Original template credits: Samuel Zurowski and Charles Barone. Adapted for McMaster SFWRENG by Dr. Istvan David.

1 Specifying a Test Plan

Following is a test plan for the `Velocity` class. It covers the intended behaviour of the class, the need for review, and testing objectives at the specification, implementation, and interaction levels.

Plan Item	**Description**
Objectives for the class	The `Velocity` class must represent the motion of the puck using a non-negative scalar speed, a direction (in degrees), and corresponding horizontal and vertical components. These values must remain internally consistent at all times. Specifically, the class should correctly construct a velocity from a given speed and direction, maintain consistency between `speed`, `speedX`, and `speedY`, and ensure that the direction determines the correct sign of each component (negative when moving left or down). In addition, bounce operations such as `reverse`, `reverseX`, and `reverseY` must update the velocity in a way that reflects expected physical behaviour. Getter methods must always return the current valid state.
Inspection / review requirements	A design and code review is required prior to testing for several reasons: 1. Specification details are unclear. The informal description does not fully define how certain cases should be handled, such as direction values outside the range 0-359, the behaviour of the default constructor, or whether angles like 360° are equivalent to 0°. These ambiguities must be resolved before writing tests. 2. We want to ensure trigonometric and numerical correctness. The calculation of `speedX` and `speedY` depends on sine and cosine functions followed by conversion to integers. This introduces potential rounding and sign errors, especially at boundary angles such as 0°, 90°, 180°, and 270°. 3. We want to ensure the preservation of internal relationships. Any method that updates the velocity must maintain consistency between the scalar speed and its components. A review helps ensure that this relationship is re-established after every update. Without this review, test cases may reflect incorrect assumptions rather than actual intended behaviour.

Plan Item	Description
Specification-based testing objectives	When treating the class as a black box, the goal is to verify that each public method behaves according to its expected contract. Testing should include: • Typical cases. Representative speeds and directions across all four quadrants. • Boundary values. Directions such as 0, 90, 180, 270, and 359, and speeds such as 0 and large values. • Equivalence classes. Combinations of positive and negative values for `speedX` and `speedY`. • Invalid inputs (if applicable). For example, negative speeds or out-of-range directions. Tests should confirm that outputs match expectations and that the observable behaviour of the class aligns with its specification.
Implementation-based testing objectives	When treating the class as a white box, the goal is to exercise the internal logic and achieve strong structural coverage. This includes: • Statement and branch coverage, ensuring all methods and decision points are executed. • Condition coverage, testing how sign decisions are made based on direction. • Path coverage, particularly for direction normalization logic. • Rounding behaviour, verifying correctness near angles where integer rounding affects results (45°, 135°, 225°, 315°). These tests increase confidence that the implementation correctly handles all internal cases.

Plan Item	Description
Interaction-based testing objectives	The goal is to verify that sequences of method calls produce consistent and expected results, and that operations compose correctly. Key interaction checks include: • `setDirection` updates components without changing speed. • `setSpeed` rescales components while preserving direction. • Applying `reverseX` and `reverseY` together produces the same result as `reverse`. • Applying reverse operations twice returns the object to its original state. • Any sequence of updates followed by getters yields a state that remains internally consistent. These tests ensure that method interactions do not violate expected behaviour over time.

2 Contract-Based Specification

Boolean expressions are defined in pseudocode. The notation `old(e)` refers to the value of an expression before execution. Directions are assumed to be in degrees and within the range $[0, 360)$.

Class Invariant

```
invariant:
    speed >= 0
    && 0 <= direction < 360
    && speedX == round(speed * cos(direction))
    && speedY == round(speed * sin(direction))
    && abs(speedX) <= speed
    && abs(speedY) <= speed
    // sign constraints based on direction
    && (direction > 90 && direction < 270 ==> speedX <= 0)
    && (direction < 90 || direction > 270 ==> speedX >= 0)
    && (direction > 180 && direction < 360 ==> speedY <= 0)
    && (direction > 0 && direction < 180 ==> speedY >= 0)
```

Constructors

Velocity()

Pre: true

Post:

```
speed == 0
&& direction == 0
&& speedX == 0
&& speedY == 0
&& invariant holds
```

Velocity(Speed speed, Direction direction)

Pre:

```
speed != null && direction != null
&& speed >= 0
&& 0 <= direction < 360
```

Post:

```
this.speed == speed
&& this.direction == direction
&& this.speedX == round(speed * cos(direction))
&& this.speedY == round(speed * sin(direction))
&& invariant holds
```

Getters

getSpeed()

Pre: true

Post:

```
result == this.speed
&& result >= 0
&& object state remains unchanged
```

getSpeedX()

Pre: true

Post:

```
result == this.speedX
&& abs(result) <= this.speed
&& object state remains unchanged
```

getSpeedY()

Pre: true

Post:

```
result == this.speedY
```

```
&& abs(result) <= this.speed
&& object state remains unchanged
```

getDirection()

Pre: true

Post:

```
result == this.direction
&& 0 <= result < 360
&& object state remains unchanged
```

Setters

setSpeed(Speed speed)

Pre:

```
speed != null && speed >= 0
```

Post:

```
this.speed == speed
&& this.direction == old(this.direction)
&& this.speedX == round(speed * cos(this.direction))
&& this.speedY == round(speed * sin(this.direction))
&& invariant holds
```

setDirection(Direction direction)

Pre:

```
direction != null && 0 <= direction < 360
```

Post:

```
this.direction == direction
&& this.speed == old(this.speed)
&& this.speedX == round(this.speed * cos(direction))
&& this.speedY == round(this.speed * sin(direction))
&& invariant holds
```

Direction Reversal Methods

reverse()

Pre: true

Post:

```
this.speed == old(this.speed)
&& this.direction == (old(this.direction) + 180) mod 360
```

```
&& this.speedX == -old(this.speedX)
&& this.speedY == -old(this.speedY)
&& invariant holds
```

reverseX()

Pre: true

Post:

```
this.speed == old(this.speed)
&& this.speedX == -old(this.speedX)
&& this.speedY == old(this.speedY)
&& this.direction == (540 - old(this.direction)) mod 360
&& invariant holds
```

reverseY()

Pre: true

Post:

```
this.speed == old(this.speed)
&& this.speedY == -old(this.speedY)
&& this.speedX == old(this.speedX)
&& this.direction == (360 - old(this.direction)) mod 360
&& invariant holds
```

3 Test Cases

Each test case specifies the input state, the operation performed, and the expected outcome.

setDirection

Test 1: updating direction preserves speed and updates components

Input: Initial state: v = new Velocity$(10, 0)$ Call: v.setDirection(90)

Expected Output:

```
v.getDirection() == 90
v.getSpeed() == 10
v.getSpeedX() == 0
v.getSpeedY() == 10
class invariant holds
```

setSpeed

Test 1: updating speed preserves direction and rescales components

Input: Initial state: v = new Velocity$(5, 180)$ Call: v.setSpeed(20)

Expected Output:

```
v.getSpeed() == 20
v.getDirection() == 180
v.getSpeedX() == -20
v.getSpeedY() == 0
class invariant holds
```

reverse

Test 1: reversing motion from right to left

Input: v = new Velocity$(10, 0)$ Call: v.reverse$()$

Expected Output:

```
direction == 180
speedX == -10
speedY == 0
speed == 10
```

Test 2: reversing from northeast to southwest

Input: v = new Velocity$(14, 45)$ Call: v.reverse$()$

Expected Output:

```
direction == 225
speedX == -10
speedY == -10
speed == 14
```

Test 3: applying reverse twice restores original state

Input: v = new Velocity$(8, 123)$ Call: v.reverse$()$ twice

Expected Output:

```
Final (speed, direction, speedX, speedY)
matches the original state
```

Test 4: reversing straight downward motion

Input: v = new Velocity$(7, 270)$ Call: v.reverse$()$

Expected Output:

```
direction == 90
speedX == 0
speedY == 7
speed == 7
```

Test 5: reversing when speed is zero

Input: v = new Velocity$(0, 60)$ Call: v.reverse()

Expected Output:

```
direction == 240
speedX == 0
speedY == 0
speed == 0
```

reverseX

Test 1: bounce off vertical wall while moving right

Input: v = new Velocity$(10, 0)$ Call: v.reverseX()

Expected Output:

```
speedX == -10
speedY == 0
direction == 180
speed == 10
```

Test 2: northeast becomes northwest

Input: v_1 = new Velocity$(14, 45)$ v_2 = new Velocity$(14, 45)$ Call: v_1.reverseX()

Expected Output:

```
v_1.direction == 135
v_1.speedX == -v_2.speedX
v_1.speedY == v_2.speedY
speed == 14
```

Test 3: vertical motion remains unchanged

Input: v = new Velocity$(10, 90)$ Call: v.reverseX()

Expected Output:

```
speedX == 0
speedY == 10
direction == 90
speed == 10
```

Test 4: applying reverseX twice restores original state

Input: v = new Velocity$(12, 200)$ Call: v.reverseX() twice

Expected Output:

```
Final state identical to initial state
(speed, direction, speedX, speedY)
```

Test 5: reverseX followed by reverseY equals reverse

Input: a = new Velocity$(10, 30)$ b = new Velocity$(10, 30)$

Call: a.reverseX(); a.reverseY(); b.reverse();

Expected Output:

```
a.speed == b.speed
a.direction == b.direction
a.speedX == b.speedX
a.speedY == b.speedY
```

4 Test Automation

Please see the associated GitHub repository for the test suite. I've used Playwright as that is the tool I am most familiar with.

a)

I selected `checkboxes` and `dropdown` as the two cases from the provided list. The tests are provided in `test_checkboxes.py` and `test_dropdown.py`, respectively.

Checkboxes (`https://the-internet.herokuapp.com/checkboxes`)

This page contains two checkboxes with predefined initial states and supports basic user interactions (checking and unchecking). The test suite verifies both the default configuration and the correctness of state transitions under user actions.

Initial state, `test_initial_checkbox_states()`: Verifies that the first checkbox is initially unchecked and the second is checked, ensuring the page loads with the expected default configuration.

Check action, `test_check_first_checkbox()`: Checks that selecting an unchecked checkbox correctly updates its state to checked, confirming basic interaction behaviour.

Uncheck action, `test_uncheck_second_checkbox()`: Verifies that an initially checked checkbox can be unchecked, ensuring that state changes in both directions are supported.

Toggle consistency, `test_toggle_twice_returns_original()`: Checks that performing a check followed by an uncheck returns the checkbox to its original state, validating stability under repeated operations.

Multiple selection, `test_both_checkboxes_checked()`: Ensures that both checkboxes can be selected simultaneously, confirming that independent elements do not interfere with each other.

Dropdown (`https://the-internet.herokuapp.com/dropdown`)

This page contains a dropdown menu with a placeholder and two selectable options. The tests verify correct selection behaviour and consistency of the selected value.

Default state, `test_default_dropdown_state()`: Verifies that no option is selected by default (placeholder is active), ensuring correct initialization of the dropdown.

Select first option, `test_select_first_option()`: Checks that selecting the first option updates the dropdown value correctly.

Select second option, `test_select_second_option()`: Verifies correct behaviour when selecting the second option.

Switching options, `test_switch_between_options()`: Ensures that changing the selection from one option to another updates the state accordingly, validating transitions between valid inputs.

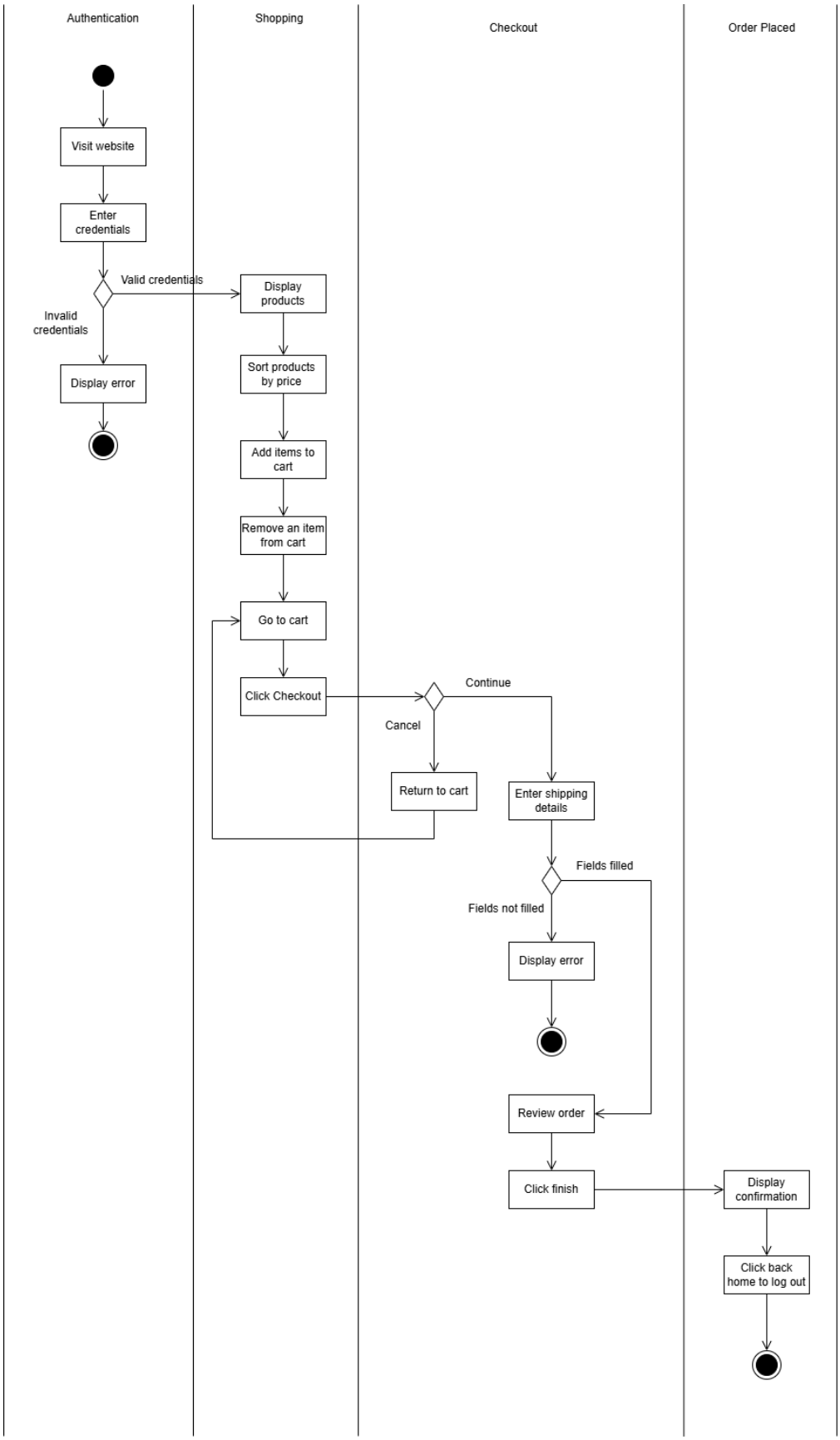
Idempotence, `test_reselect_same_option()`: Checks that selecting the same option multiple times does not produce unexpected behaviour and the state remains consistent.

The website passes all of the above tests.

```
salee@rogzephyrus MINGW64 ~/OneDrive/Documents/GitHub/3S03-A3 (main)
$ pytest test_dropdown.py -v
============================================================= test session starts =============================================================
platform win32 -- Python 3.10.6, pytest-9.0.3, pluggy-1.6.0 -- C:\Users\salee\AppData\Local\Programs\Python\Python310\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\salee\OneDrive\Documents\GitHub\3S03-A3
plugins: anyio-4.9.0
collected 5 items

test_dropdown.py::test_default_dropdown_state PASSED                                                                                    [ 20%]
test_dropdown.py::test_select_first_option PASSED                                                                                       [ 40%]
test_dropdown.py::test_select_second_option PASSED                                                                                      [ 60%]
test_dropdown.py::test_switch_between_options PASSED                                                                                    [ 80%]
test_dropdown.py::test_reselect_same_option PASSED                                                                                      [100%]

============================================================== 5 passed in 9.11s ==============================================================

salee@rogzephyrus MINGW64 ~/OneDrive/Documents/GitHub/3S03-A3 (main)
$  pytest test_checkboxes.py -v
============================================================= test session starts =============================================================
platform win32 -- Python 3.10.6, pytest-9.0.3, pluggy-1.6.0 -- C:\Users\salee\AppData\Local\Programs\Python\Python310\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\salee\OneDrive\Documents\GitHub\3S03-A3
plugins: anyio-4.9.0
collected 5 items

test_checkboxes.py::test_initial_checkbox_states PASSED                                                                                 [ 20%]
test_checkboxes.py::test_check_first_checkbox PASSED                                                                                    [ 40%]
test_checkboxes.py::test_uncheck_second_checkbox PASSED                                                                                 [ 60%]
test_checkboxes.py::test_toggle_twice_returns_original PASSED                                                                           [ 80%]
test_checkboxes.py::test_both_checkboxes_checked PASSED                                                                                 [100%]

============================================================== 5 passed in 8.86s ==============================================================

salee@rogzephyrus MINGW64 ~/OneDrive/Documents/GitHub/3S03-A3 (main)
$
```

b)

The tests are provided in `test_saucedemo.py`. An activity diagram for this process is shown below. Apologies for the image quality as the diagram is kind of long, I have also attached an svg of the same diagram in the GitHub repo in case this one is difficult to read.


Authentication
Shopping
Checkout
Order Placed
Visit website
Enter credentials
Valid credentials
Invalid credentials
Display error
Display products
Sort products by price
Add items to cart
Remove an item from cart
Go to cart
Click Checkout
Continue
Cancel
Return to cart
Enter shipping details
Fields filled
Fields not filled
Display error
Review order
Click finish
Display confirmation
Click back home to log out

Rationale

I first mapped out the user flow using an activity diagram, splitting it into four main parts: authentication, shopping, checkout, and order placed. This made it easier to see where the system branches (like valid vs invalid login, cancel vs continue at checkout, and whether required fields are filled or not).

The main test (`test_complete_purchase_flow`) follows the full happy path. This includes logging in successfully, viewing and sorting products, adding and removing items, going through checkout, verifying totals, and finishing the order. This test basically ensures the core functionality works end-to-end.

To cover the important branches in the diagram, I added a few smaller targeted tests:

- `test_invalid_user_displays_error`: checks the invalid login case.
- `test_checkout_with_missing_fields_displays_error`: checks that the form validation triggers when required fields are missing.
- `test_cancel_checkout_returns_to_cart`: checks that cancelling checkout correctly returns the user to the cart.

So overall, each decision point in the diagram is covered at least once. The happy path handles all the "successful" transitions, and the extra tests cover the main failure/alternative paths.

I didn't try to test every possible invalid input (like every missing field combination or all bad credential cases), since those would mostly hit the same validation logic and wouldn't add much value. Instead, I focused on cases that actually represent different behaviours in the system.

I also included a more meaningful check during checkout (verifying that subtotal + tax equals the total), so it's not just clicking through pages. Finally, using Playwright's `expect()` helps avoid flaky tests since it waits for the UI state instead of relying on timing.